

# ANJANA KRISHNAMOORTHY

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## PROFESSIONAL SUMMARY

Software Engineer with expertise in **Python, Next.js, PostgreSQL, and REST API design**, plus applied experience in **ML systems and data pipelines**. Skilled in scalable backend architecture, CI/CD, and reliable production delivery. Backed by published research on multi-agent reinforcement learning and a biomedical retrieval system under peer review.

## EDUCATION

**California State University, Dominguez Hills** Carson, CA  
*Master of Science in Computer Science, GPA: 3.9* Aug 2024 – May 2026  
• Coursework: Algorithms, Operating Systems, Database Management Systems, Object-Oriented Design

**CSI College of Engineering** India  
*Bachelor of Science in Computer Science* 2019 – 2023

## PROFESSIONAL EXPERIENCE

**Data Analyst** Mar 2024 – Apr 2024  
*Medtourney* India  
• Built and deployed a **logistic regression** mortality-prediction model on 50K+ clinical records, achieving **98% accuracy**; engineered automated **ETL** pipelines in Python to streamline data ingestion and analysis.

**Python Developer** Feb 2024 – Mar 2024  
*Oasis Infobyte* Remote  
• Engineered automated **Python** scripts to replace manual data-processing workflows, cutting processing time by **30%** and improving reporting speed and accuracy.

**Associate Statistical Data Scientist Intern** Apr 2023 – Jul 2023  
*Pfizer* India  
• Analyzed large-scale clinical trial datasets in **SAS** with 99%+ accuracy against FDA compliance standards; re-engineered SDTM/ADaM reporting workflows, cutting TLF generation time by **35%**.

## ADDITIONAL EXPERIENCE

**Fab Lab & Software Technician** Feb 2025 – May 2026  
*Toro Auxiliary Partners – CSUDH* Carson, CA  
• Delivered **Swift** and **Python** programming workshops to 100+ K–12 students using Sphero Edu robotics.

## PROJECTS

**Timesheet Management Application** | *Python, Next.js, PostgreSQL, REST APIs, CI/CD* Feb 2026  
• Designed and led system architecture for a timesheet application supporting **50+ contractors and staff**: normalized **PostgreSQL** schema, RESTful API decoupled from a **Next.js** client, and transaction isolation for concurrent multi-user writes.  
• Built and maintained **CI/CD** pipelines (GitHub Actions) with automated test gates and staged rollouts, maintaining zero production defects to date.

**Adaptive Temporal Multi-Model RAG** | *Python, BioBERT, FAISS, BM25, DeBERTa-v3, Gradio* May 2026  
• Architected a biomedical QA system with hybrid dense (FAISS) + lexical (BM25) retrieval, domain routing, and temporal ranking; indexed 8,790 documents from PubMed, Europe PMC, and Medical QnA using **BioBERT** embeddings, deployed as a Gradio web app.  
• Achieved **94.6% silver-label accuracy**, 87.9% evidence relevance, and 96.7% recent-evidence retrieval on a 240-question benchmark; integrated **DeBERTa-v3** for claim-level NLI verification.

**FINDIT – Offline Android SMS Assistant** | *Android, Java, SQLite, Jetpack Compose, SHA-256* Dec 2025  
• Built an offline-first Android assistant with 6+ command-driven features; implemented **SHA-256** SMS command authentication for secure-by-design request handling with no internet dependency.  
• Optimized **SQLite** query patterns to reduce data retrieval time by 40% across 10+ test devices.

## RESEARCH & PUBLICATION

**Multi-Agent RL for Interactive Code Debugging with Human Feedback and Memory** | *RANLP 2025* Sept 2025  
• Coauthored a peer-reviewed paper on RLHF-based multi-agent PPO and BERT architecture with FAISS-powered memory retrieval, achieving **21% higher bug resolution accuracy** versus baseline models.

## SKILLS

**Languages:** Python, Java, TypeScript, JavaScript, SQL, C, SAS  
**AI & ML:** RAG, BioBERT, DeBERTa, FAISS, BM25, Deep Learning, Computer Vision, LLM Tools, AI Agent Integration  
**Backend & Mobile:** REST APIs, Next.js, Node.js, Android (Java), Jetpack Compose, SQLite, SOLID Principles, Design Patterns  
**Cloud & DevOps:** AWS (EC2, S3, Lambda, CloudWatch, IAM), Azure, Docker, Jenkins, CI/CD, Git/GitHub  
**Data & Testing:** PostgreSQL, MySQL, Oracle, ETL, SDTM/ADaM, Pytest, Jest, Selenium, Postman, Pandas, Scikit-learn